安裝 tiebreaker 軟體套件

ONTAP MetroCluster

NetApp
March 22, 2024

This PDF was generated from https://docs.netapp.com/zh-tw/ontap-metrocluster/tiebreaker/install-choose-procedure.html on March 22, 2024. Always check docs.netapp.com for the latest.

目錄

安裝 tiebreaker 軟體套件 1
選擇您的安裝程序 1
安裝斷路器 1.6 1
安裝斷路器 1.5 或 1.4 20

安裝 tiebreaker 軟體套件

選擇您的安裝程序

您遵循的 tiebreaker 安裝程序取決於您要安裝的 tiebreaker 版本。

Tiebreaker 版本	前往...
斷路器 1.6.	"安裝斷路器 1.6"
斷路器 1.5 或 1.4	"安裝斷路器 1.5 或 1.4"

安裝斷路器 1.6

執行新的安裝或升級 tiebreaker 軟體。

關於這項工作

- 您的儲存系統必須執行 ONTAP 9.12.1 或更新版本。
- 您可以將 MetroCluster tiebreaker 安裝為非 root 使用者、並擁有足夠的管理權限來執行 tiebreaker 安裝、建立表格和使用者、以及設定使用者密碼。

步驟

1. 下載 MetroCluster Tiebreaker 1.6 軟體。

   "NetApp 支援網站"

2. 以root使用者身分登入主機。
3. 如果您正在執行升級、請確認您正在執行的 tiebreaker 版本：

   以下範例顯示 tiebreaker 1.5 。

   ```
   [root@mcctb ~] # netapp-metrocluster-tiebreaker-software-cli
   NetApp MetroCluster Tiebreaker :> version show
   NetApp MetroCluster Tiebreaker 1.5: Sun Mar 13 09:59:02 IST 2022
   NetApp MetroCluster Tiebreaker :> exit
   ```

4. 安裝或升級 tiebreaker 軟體。

安裝斷路器 **1.6**

請依照下列步驟安裝斷路器 1.6 。

步驟

a. 在執行下列命令 `[root@mcctb ~] #` 開始安裝的提示：

```
sh MetroClusterTiebreakerInstall-1.6
```

系統會顯示下列輸出、以利成功安裝：

```
Extracting the MetroCluster Tiebreaker installation/upgrade
archive
Install digest hash is Ok
Performing the MetroCluster Tiebreaker code signature check
Install code signature is Ok
Enter unix user account to use for the installation:
mcctbadminuser
Unix user account "mcctbadminuser" doesn't exist. Do you wish to
create "mcctbadminuser" user account? [Y/N]: y
useradd: warning: the home directory already exists.
Not copying any file from skel directory into it.
Creating mailbox file: File exists
Unix account "mcctbadminuser" created.
Changing password for user mcctbadminuser.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
MetroCluster Tiebreaker requires unix user account
"mcctbadminuser" to be added to the group "mcctbgrp" for admin
access.
Do you wish to add ? [Y/N]: y
Unix user account "mcctbadminuser" added to "mcctbgrp".
Do you wish to generate your own public-private key pair for
encrypting audit log? [Y/N]: y
Generating public-private key pair...
Configuring Vault...
Starting vault server...
==> Vault server configuration:

             Api Address: <api_address>
                     Cgo: disabled
         Cluster Address: <cluster_address>
   Environment Variables: BASH_FUNC_which%%,
DBUS_SESSION_BUS_ADDRESS, GODEBUG, HISTCONTROL, HISTSIZE, HOME,
```

```
HOSTNAME, HOST_ACCOUNT, LANG, LESSOPEN, LOGNAME, LS_COLORS, MAIL,
PATH, PWD, SHELL, SHLVL, SSH_CLIENT, SSH_CONNECTION, SSH_TTY,
STAF_TEMP_DIR, TERM, USER, VAULT_ADDR, VAULT_TOKEN,
XDG_RUNTIME_DIR, XDG_SESSION_ID, _, vault_Addr, which_declare
              Go Version: go1.20.5
              Listener 1: tcp (addr: "0.0.0.0:8200", cluster
address: "0.0.0.0:8201", max_request_duration: "1m30s",
max_request_size: "33554432", tls: "enabled")
               Log Level:
                   Mlock: supported: true, enabled: true
           Recovery Mode: false
                 Storage: file
                 Version: Vault v1.14.0, built 2023-06-
19T11:40:23Z
             Version Sha:
13a649f860186dffe3f3a4459814d87191efc321

==> Vault server started! Log data will stream in below:

2023-11-23T15:14:28.532+0530 [INFO]  proxy environment:
http_proxy="" https_proxy="" no_proxy=""
2023-11-23T15:14:28.577+0530 [INFO]  core: Initializing version
history cache for core
2023-11-23T15:14:38.552+0530 [INFO]  core: security barrier not
initialized
2023-11-23T15:14:38.552+0530 [INFO]  core: seal configuration
missing, not initialized
2023-11-23T15:14:38.554+0530 [INFO]  core: security barrier not
initialized
2023-11-23T15:14:38.555+0530 [INFO]  core: security barrier
initialized: stored=1 shares=5 threshold=3
2023-11-23T15:14:38.556+0530 [INFO]  core: post-unseal setup
starting
2023-11-23T15:14:38.577+0530 [INFO]  core: loaded wrapping token
key
2023-11-23T15:14:38.577+0530 [INFO]  core: successfully setup
plugin catalog: plugin-directory=""
2023-11-23T15:14:38.577+0530 [INFO]  core: no mounts; adding
default mount table
2023-11-23T15:14:38.578+0530 [INFO]  core: successfully mounted:
type=cubbyhole version="v1.14.0+builtin.vault" path=cubbyhole/
namespace="ID: root. Path: "
2023-11-23T15:14:38.578+0530 [INFO]  core: successfully mounted:
type=system version="v1.14.0+builtin.vault" path=sys/
namespace="ID: root. Path: "
2023-11-23T15:14:38.578+0530 [INFO]  core: successfully mounted:
```

```
type=identity version="v1.14.0+builtin.vault" path=identity/
namespace="ID: root. Path: "
2023-11-23T15:14:38.581+0530 [INFO]  core: successfully mounted:
type=token version="v1.14.0+builtin.vault" path=token/
namespace="ID: root. Path: "
2023-11-23T15:14:38.581+0530 [INFO]  rollback: starting rollback
manager
2023-11-23T15:14:38.581+0530 [INFO]  core: restoring leases
2023-11-23T15:14:38.582+0530 [INFO]  expiration: lease restore
complete
2023-11-23T15:14:38.582+0530 [INFO]  identity: entities restored
2023-11-23T15:14:38.582+0530 [INFO]  identity: groups restored
2023-11-23T15:14:38.583+0530 [INFO]  core: Recorded vault
version: vault version=1.14.0 upgrade time="2023-11-23
09:44:38.582881162 +0000 UTC" build date=2023-06-19T11:40:23Z
2023-11-23T15:14:38.583+0530 [INFO]  core: usage gauge collection
is disabled
2023-11-23T15:14:38.998+0530 [INFO]  core: post-unseal setup
complete
2023-11-23T15:14:38.999+0530 [INFO]  core: root token generated
2023-11-23T15:14:38.999+0530 [INFO]  core: pre-seal teardown
starting
2023-11-23T15:14:38.999+0530 [INFO]  rollback: stopping rollback
manager
2023-11-23T15:14:38.999+0530 [INFO]  core: pre-seal teardown
complete
2023-11-23T15:14:39.311+0530 [INFO]  core.cluster-listener.tcp:
starting listener: listener_address=0.0.0.0:8201
2023-11-23T15:14:39.311+0530 [INFO]  core.cluster-listener:
serving cluster requests: cluster_listen_address=[::]:8201
2023-11-23T15:14:39.312+0530 [INFO]  core: post-unseal setup
starting
2023-11-23T15:14:39.312+0530 [INFO]  core: loaded wrapping token
key
2023-11-23T15:14:39.312+0530 [INFO]  core: successfully setup
plugin catalog: plugin-directory=""
2023-11-23T15:14:39.313+0530 [INFO]  core: successfully mounted:
type=system version="v1.14.0+builtin.vault" path=sys/
namespace="ID: root. Path: "
2023-11-23T15:14:39.313+0530 [INFO]  core: successfully mounted:
type=identity version="v1.14.0+builtin.vault" path=identity/
namespace="ID: root. Path: "
2023-11-23T15:14:39.313+0530 [INFO]  core: successfully mounted:
type=cubbyhole version="v1.14.0+builtin.vault" path=cubbyhole/
namespace="ID: root. Path: "
2023-11-23T15:14:39.314+0530 [INFO]  core: successfully mounted:
```

```
type=token version="v1.14.0+builtin.vault" path=token/
namespace="ID: root. Path: "
2023-11-23T15:14:39.314+0530 [INFO]  rollback: starting rollback
manager
2023-11-23T15:14:39.314+0530 [INFO]  core: restoring leases
2023-11-23T15:14:39.314+0530 [INFO]  identity: entities restored
2023-11-23T15:14:39.314+0530 [INFO]  expiration: lease restore
complete
2023-11-23T15:14:39.314+0530 [INFO]  identity: groups restored
2023-11-23T15:14:39.315+0530 [INFO]  core: usage gauge collection
is disabled
2023-11-23T15:14:39.316+0530 [INFO]  core: post-unseal setup
complete
2023-11-23T15:14:39.316+0530 [INFO]  core: vault is unsealed
Success! Uploaded policy: mcctb-policy
2023-11-23T15:14:39.795+0530 [INFO]  core: enabled credential
backend: path=approle/ type=approle version=""
Success! Enabled approle auth method at: approle/
2023-11-23T15:14:39.885+0530 [INFO]  core: successful mount:
namespace="" path=mcctb/ type=kv version=""
Success! Enabled the kv secrets engine at: mcctb/
Success! Data written to: auth/approle/role/mcctb-app
Installing the NetApp-MetroCluster-Tiebreaker-Software-1.6-
1.x86_64.rpm
Preparing...                          #
################################# # [100%]

Updating / installing...

1:NetApp-MetroCluster-Tiebreaker-So#
################################# # [100%]
Performing file integrity check
etc/cron.weekly/metrocluster-tiebreaker-support is Ok
etc/cron.weekly/metrocluster-tiebreaker-support-cov is Ok
etc/init.d/netapp-metrocluster-tiebreaker-software is Ok
etc/init.d/netapp-metrocluster-tiebreaker-software-cov is Ok
etc/logrotate.d/mcctb is Ok
opt/netapp/mcctb/lib/common/activation-1.1.1.jar is Ok
opt/netapp/mcctb/lib/common/aopalliance.jar is Ok
opt/netapp/mcctb/lib/common/args4j.jar is Ok
opt/netapp/mcctb/lib/common/aspectjrt.jar is Ok
opt/netapp/mcctb/lib/common/aspectjweaver.jar is Ok
opt/netapp/mcctb/lib/common/asup.jar is Ok
opt/netapp/mcctb/lib/common/bcpkix-jdk15on.jar is Ok
opt/netapp/mcctb/lib/common/bcprov-jdk15on.jar is Ok
opt/netapp/mcctb/lib/common/bcprov-jdk18on.jar is Ok
```

```
opt/netapp/mcctb/lib/common/bctls-fips-1.0.13.jar is Ok
opt/netapp/mcctb/lib/common/bctls-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/bcutil-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/cglib.jar is Ok
opt/netapp/mcctb/lib/common/commons-codec.jar is Ok
opt/netapp/mcctb/lib/common/commons-collections4.jar is Ok
opt/netapp/mcctb/lib/common/commons-compress.jar is Ok
opt/netapp/mcctb/lib/common/commons-daemon.jar is Ok
opt/netapp/mcctb/lib/common/commons-daemon.src.jar is Ok
opt/netapp/mcctb/lib/common/commons-dbcp2.jar is Ok
opt/netapp/mcctb/lib/common/commons-io.jar is Ok
opt/netapp/mcctb/lib/common/commons-lang3.jar is Ok
opt/netapp/mcctb/lib/common/commons-logging.jar is Ok
opt/netapp/mcctb/lib/common/commons-pool2.jar is Ok
opt/netapp/mcctb/lib/common/guava.jar is Ok
opt/netapp/mcctb/lib/common/httpclient.jar is Ok
opt/netapp/mcctb/lib/common/httpcore.jar is Ok
opt/netapp/mcctb/lib/common/jakarta.activation.jar is Ok
opt/netapp/mcctb/lib/common/jakarta.xml.bind-api.jar is Ok
opt/netapp/mcctb/lib/common/java-xmlbuilder.jar is Ok
opt/netapp/mcctb/lib/common/javax.inject.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-api-2.3.1.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-core.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-impl.jar is Ok
opt/netapp/mcctb/lib/common/jline.jar is Ok
opt/netapp/mcctb/lib/common/jna.jar is Ok
opt/netapp/mcctb/lib/common/joda-time.jar is Ok
opt/netapp/mcctb/lib/common/jsch.jar is Ok
opt/netapp/mcctb/lib/common/json.jar is Ok
opt/netapp/mcctb/lib/common/jsvc.zip is Ok
opt/netapp/mcctb/lib/common/junixsocket-common.jar is Ok
opt/netapp/mcctb/lib/common/junixsocket-native-common.jar is Ok
opt/netapp/mcctb/lib/common/logback-classic.jar is Ok
opt/netapp/mcctb/lib/common/logback-core.jar is Ok
opt/netapp/mcctb/lib/common/mail-1.6.2.jar is Ok
opt/netapp/mcctb/lib/common/mariadb-java-client.jar is Ok
opt/netapp/mcctb/lib/common/mcctb-mib.jar is Ok
opt/netapp/mcctb/lib/common/mcctb.jar is Ok
opt/netapp/mcctb/lib/common/mockito-core.jar is Ok
opt/netapp/mcctb/lib/common/slf4j-api.jar is Ok
opt/netapp/mcctb/lib/common/snmp4j.jar is Ok
opt/netapp/mcctb/lib/common/spring-aop.jar is Ok
opt/netapp/mcctb/lib/common/spring-beans.jar is Ok
opt/netapp/mcctb/lib/common/spring-context-support.jar is Ok
opt/netapp/mcctb/lib/common/spring-context.jar is Ok
opt/netapp/mcctb/lib/common/spring-core.jar is Ok
```

```
opt/netapp/mcctb/lib/common/spring-expression.jar is Ok
opt/netapp/mcctb/lib/common/spring-web.jar is Ok
opt/netapp/mcctb/lib/common/vault-java-driver.jar is Ok
opt/netapp/mcctb/lib/common/xz.jar is Ok
opt/netapp/mcctb/lib/org.jacoco.agent-0.8.8-runtime.jar is Ok
opt/netapp/mcctb/bin/mcctb-asup-invoke is Ok
opt/netapp/mcctb/bin/mcctb_postrotate is Ok
opt/netapp/mcctb/bin/netapp-metrocluster-tiebreaker-software-cli
is Ok
/

Synchronizing state of netapp-metrocluster-tiebreaker-
software.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable netapp-
metrocluster-tiebreaker-software
Created symlink /etc/systemd/system/multi-
user.target.wants/netapp-metrocluster-tiebreaker-software.service
→ /etc/systemd/system/netapp-metrocluster-tiebreaker-
software.service.

Attempting to start NetApp MetroCluster Tiebreaker software
services
Started NetApp MetroCluster Tiebreaker software services
Successfully installed NetApp MetroCluster Tiebreaker software
version 1.6.


IMPORTANT: Please ensure you store the following private key used
to decrypt audit logs securely in a safe location.
Consider the following recommendations:
- Save it to an encrypted storage device or file.
- Use strong, unique passwords to protect the storage device or
file.
- Restrict access to the private key file to authorized
individuals only.

The Private key is:

-----BEGIN PRIVATE KEY-----

<private_key_id>

-----END PRIVATE KEY-----
```

從 tiebreaker 1.5 升級至 1.6

請依照下列步驟、將 tiebreaker 1.5 軟體版本升級至 tiebreaker 1.6 。

步驟

a. 在執行下列命令 `[root@mcctb ~] #` 提示升級軟體：

```
sh MetroClusterTiebreakerInstall-1.6
```

系統會顯示下列輸出以成功升級：

```
Extracting the MetroCluster Tiebreaker installation/upgrade
archive
Install digest hash is Ok
Performing the MetroCluster Tiebreaker code signature check
Install code signature is Ok

Enter database user name : root

Please enter database password for root
Enter password:

Password updated successfully in the database.

Do you wish to generate your own public-private key pair for
encrypting audit log? [Y/N]: y
Generating public-private key pair...
Configuring Vault...
==> Vault shutdown triggered
2023-07-21T00:30:22.335+0530 [INFO]  core: marked as sealed
2023-07-21T00:30:22.335+0530 [INFO]  core: pre-seal teardown
starting
2023-07-21T00:30:22.335+0530 [INFO]  rollback: stopping rollback
manager
2023-07-21T00:30:22.335+0530 [INFO]  core: pre-seal teardown
complete
2023-07-21T00:30:22.335+0530 [INFO]  core: stopping cluster
listeners
2023-07-21T00:30:22.335+0530 [INFO]  core.cluster-listener:
forwarding rpc listeners stopped
2023-07-21T00:30:22.375+0530 [INFO]  core.cluster-listener: rpc
listeners successfully shut down
2023-07-21T00:30:22.375+0530 [INFO]  core: cluster listeners
successfully shut down
2023-07-21T00:30:22.376+0530 [INFO]  core: vault is sealed
Starting vault server...
==> Vault server configuration:
```

```
             Api Address: <api_address>
                     Cgo: disabled
         Cluster Address: <cluster_address>
   Environment Variables: BASH_FUNC_which%%,
DBUS_SESSION_BUS_ADDRESS, GODEBUG, HISTCONTROL, HISTSIZE, HOME,
HOSTNAME, HOST_ACCOUNT, LANG, LESSOPEN, LOGNAME, LS_COLORS, MAIL,
PATH, PWD, SHELL, SHLVL, SSH_CLIENT, SSH_CONNECTION, SSH_TTY,
STAF_TEMP_DIR, TERM, USER, VAULT_ADDR, VAULT_TOKEN,
XDG_RUNTIME_DIR, XDG_SESSION_ID, _, vault_Addr, which_declare
              Go Version: go1.20.5
              Listener 1: tcp (addr: "0.0.0.0:8200", cluster
address: "0.0.0.0:8201", max_request_duration: "1m30s",
max_request_size: "33554432", tls: "enabled")
               Log Level:
                   Mlock: supported: true, enabled: true
           Recovery Mode: false
                 Storage: file
                 Version: Vault v1.14.0, built 2023-06-
19T11:40:23Z
             Version Sha:
13a649f860186dffe3f3a4459814d87191efc321

==> Vault server started! Log data will stream in below:

2023-07-21T00:30:33.065+0530 [INFO]  proxy environment:
http_proxy="" https_proxy="" no_proxy=""
2023-07-21T00:30:33.098+0530 [INFO]  core: Initializing version
history cache for core
2023-07-21T00:30:43.092+0530 [INFO]  core: security barrier not
initialized
2023-07-21T00:30:43.092+0530 [INFO]  core: seal configuration
missing, not initialized
2023-07-21T00:30:43.094+0530 [INFO]  core: security barrier not
initialized
2023-07-21T00:30:43.096+0530 [INFO]  core: security barrier
initialized: stored=1 shares=5 threshold=3
2023-07-21T00:30:43.098+0530 [INFO]  core: post-unseal setup
starting
2023-07-21T00:30:43.124+0530 [INFO]  core: loaded wrapping token
key
2023-07-21T00:30:43.124+0530 [INFO]  core: successfully setup
plugin catalog: plugin-directory=""
2023-07-21T00:30:43.124+0530 [INFO]  core: no mounts; adding
default mount table
2023-07-21T00:30:43.125+0530 [INFO]  core: successfully mounted:
```

```
type=cubbyhole version="v1.14.0+builtin.vault" path=cubbyhole/
namespace="ID: root. Path: "
2023-07-21T00:30:43.126+0530 [INFO]  core: successfully mounted:
type=system version="v1.14.0+builtin.vault" path=sys/
namespace="ID: root. Path: "
2023-07-21T00:30:43.126+0530 [INFO]  core: successfully mounted:
type=identity version="v1.14.0+builtin.vault" path=identity/
namespace="ID: root. Path: "
2023-07-21T00:30:43.129+0530 [INFO]  core: successfully mounted:
type=token version="v1.14.0+builtin.vault" path=token/
namespace="ID: root. Path: "
2023-07-21T00:30:43.130+0530 [INFO]  rollback: starting rollback
manager
2023-07-21T00:30:43.130+0530 [INFO]  core: restoring leases
2023-07-21T00:30:43.130+0530 [INFO]  identity: entities restored
2023-07-21T00:30:43.130+0530 [INFO]  identity: groups restored
2023-07-21T00:30:43.131+0530 [INFO]  core: usage gauge collection
is disabled
2023-07-21T00:30:43.131+0530 [INFO]  expiration: lease restore
complete
2023-07-21T00:30:43.131+0530 [INFO]  core: Recorded vault
version: vault version=1.14.0 upgrade time="2023-07-20
19:00:43.131158543 +0000 UTC" build date=2023-06-19T11:40:23Z
2023-07-21T00:30:43.371+0530 [INFO]  core: post-unseal setup
complete
2023-07-21T00:30:43.371+0530 [INFO]  core: root token generated
2023-07-21T00:30:43.371+0530 [INFO]  core: pre-seal teardown
starting
2023-07-21T00:30:43.371+0530 [INFO]  rollback: stopping rollback
manager
2023-07-21T00:30:43.372+0530 [INFO]  core: pre-seal teardown
complete
2023-07-21T00:30:43.694+0530 [INFO]  core.cluster-listener.tcp:
starting listener: listener_address=0.0.0.0:8201
2023-07-21T00:30:43.695+0530 [INFO]  core.cluster-listener:
serving cluster requests: cluster_listen_address=[::]:8201
2023-07-21T00:30:43.695+0530 [INFO]  core: post-unseal setup
starting
2023-07-21T00:30:43.696+0530 [INFO]  core: loaded wrapping token
key
2023-07-21T00:30:43.696+0530 [INFO]  core: successfully setup
plugin catalog: plugin-directory=""
2023-07-21T00:30:43.697+0530 [INFO]  core: successfully mounted:
type=system version="v1.14.0+builtin.vault" path=sys/
namespace="ID: root. Path: "
2023-07-21T00:30:43.698+0530 [INFO]  core: successfully mounted:
```

```
type=identity version="v1.14.0+builtin.vault" path=identity/
namespace="ID: root. Path: "
2023-07-21T00:30:43.698+0530 [INFO]  core: successfully mounted:
type=cubbyhole version="v1.14.0+builtin.vault" path=cubbyhole/
namespace="ID: root. Path: "
2023-07-21T00:30:43.701+0530 [INFO]  core: successfully mounted:
type=token version="v1.14.0+builtin.vault" path=token/
namespace="ID: root. Path: "
2023-07-21T00:30:43.701+0530 [INFO]  rollback: starting rollback
manager
2023-07-21T00:30:43.702+0530 [INFO]  core: restoring leases
2023-07-21T00:30:43.702+0530 [INFO]  identity: entities restored
2023-07-21T00:30:43.702+0530 [INFO]  expiration: lease restore
complete
2023-07-21T00:30:43.702+0530 [INFO]  identity: groups restored
2023-07-21T00:30:43.702+0530 [INFO]  core: usage gauge collection
is disabled
2023-07-21T00:30:43.703+0530 [INFO]  core: post-unseal setup
complete
2023-07-21T00:30:43.703+0530 [INFO]  core: vault is unsealed
Success! Uploaded policy: mcctb-policy
2023-07-21T00:30:44.226+0530 [INFO]  core: enabled credential
backend: path=approle/ type=approle version=""
Success! Enabled approle auth method at: approle/
2023-07-21T00:30:44.315+0530 [INFO]  core: successful mount:
namespace="" path=mcctb/ type=kv version=""
Success! Enabled the kv secrets engine at: mcctb/
Success! Data written to: auth/approle/role/mcctb-app
Upgrading to NetApp-MetroCluster-Tiebreaker-Software-1.6-
1.x86_64.rpm
Preparing...
################################# [100%]
Updating / installing...
   1:NetApp-MetroCluster-Tiebreaker-
So################################# [ 50%]
Performing file integrity check
etc/cron.weekly/metrocluster-tiebreaker-support is Ok
etc/cron.weekly/metrocluster-tiebreaker-support-cov is Ok
etc/init.d/netapp-metrocluster-tiebreaker-software is Ok
etc/init.d/netapp-metrocluster-tiebreaker-software-cov is Ok
etc/logrotate.d/mcctb is Ok
opt/netapp/mcctb/lib/common/activation-1.1.1.jar is Ok
opt/netapp/mcctb/lib/common/aopalliance.jar is Ok
opt/netapp/mcctb/lib/common/args4j.jar is Ok
opt/netapp/mcctb/lib/common/aspectjrt.jar is Ok
opt/netapp/mcctb/lib/common/aspectjweaver.jar is Ok
```

```
opt/netapp/mcctb/lib/common/asup.jar is Ok
opt/netapp/mcctb/lib/common/bcpkix-jdk15on.jar is Ok
opt/netapp/mcctb/lib/common/bcprov-jdk15on.jar is Ok
opt/netapp/mcctb/lib/common/bcprov-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/bctls-fips-1.0.13.jar is Ok
opt/netapp/mcctb/lib/common/bctls-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/bcutil-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/cglib.jar is Ok
opt/netapp/mcctb/lib/common/commons-codec.jar is Ok
opt/netapp/mcctb/lib/common/commons-collections4.jar is Ok
opt/netapp/mcctb/lib/common/commons-compress.jar is Ok
opt/netapp/mcctb/lib/common/commons-daemon.jar is Ok
opt/netapp/mcctb/lib/common/commons-daemon.src.jar is Ok
opt/netapp/mcctb/lib/common/commons-dbcp2.jar is Ok
opt/netapp/mcctb/lib/common/commons-io.jar is Ok
opt/netapp/mcctb/lib/common/commons-lang3.jar is Ok
opt/netapp/mcctb/lib/common/commons-logging.jar is Ok
opt/netapp/mcctb/lib/common/commons-pool2.jar is Ok
opt/netapp/mcctb/lib/common/guava.jar is Ok
opt/netapp/mcctb/lib/common/httpclient.jar is Ok
opt/netapp/mcctb/lib/common/httpcore.jar is Ok
opt/netapp/mcctb/lib/common/jakarta.activation.jar is Ok
opt/netapp/mcctb/lib/common/jakarta.xml.bind-api.jar is Ok
opt/netapp/mcctb/lib/common/java-xmlbuilder.jar is Ok
opt/netapp/mcctb/lib/common/javax.inject.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-api-2.3.1.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-core.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-impl.jar is Ok
opt/netapp/mcctb/lib/common/jline.jar is Ok
opt/netapp/mcctb/lib/common/jna.jar is Ok
opt/netapp/mcctb/lib/common/joda-time.jar is Ok
opt/netapp/mcctb/lib/common/jsch.jar is Ok
opt/netapp/mcctb/lib/common/json.jar is Ok
opt/netapp/mcctb/lib/common/jsvc.zip is Ok
opt/netapp/mcctb/lib/common/junixsocket-common.jar is Ok
opt/netapp/mcctb/lib/common/junixsocket-native-common.jar is Ok
opt/netapp/mcctb/lib/common/logback-classic.jar is Ok
opt/netapp/mcctb/lib/common/logback-core.jar is Ok
opt/netapp/mcctb/lib/common/mail-1.6.2.jar is Ok
opt/netapp/mcctb/lib/common/mariadb-java-client.jar is Ok
opt/netapp/mcctb/lib/common/mcctb-mib.jar is Ok
opt/netapp/mcctb/lib/common/mcctb.jar is Ok
opt/netapp/mcctb/lib/common/mockito-core.jar is Ok
opt/netapp/mcctb/lib/common/slf4j-api.jar is Ok
opt/netapp/mcctb/lib/common/snmp4j.jar is Ok
opt/netapp/mcctb/lib/common/spring-aop.jar is Ok
```

```
opt/netapp/mcctb/lib/common/spring-beans.jar is Ok
opt/netapp/mcctb/lib/common/spring-context-support.jar is Ok
opt/netapp/mcctb/lib/common/spring-context.jar is Ok
opt/netapp/mcctb/lib/common/spring-core.jar is Ok
opt/netapp/mcctb/lib/common/spring-expression.jar is Ok
opt/netapp/mcctb/lib/common/spring-web.jar is Ok
opt/netapp/mcctb/lib/common/vault-java-driver.jar is Ok
opt/netapp/mcctb/lib/common/xz.jar is Ok
opt/netapp/mcctb/bin/mcctb_postrotate is Ok
opt/netapp/mcctb/bin/netapp-metrocluster-tiebreaker-software-cli
is Ok
/

Synchronizing state of netapp-metrocluster-tiebreaker-
software.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable netapp-
metrocluster-tiebreaker-software

Attempting to start NetApp MetroCluster Tiebreaker software
services
Started NetApp MetroCluster Tiebreaker software services
Successfully upgraded NetApp MetroCluster Tiebreaker software to
version 1.6.
Cleaning up / removing...
   2:NetApp-MetroCluster-Tiebreaker-
So################################ [100%]


IMPORTANT: Please ensure you store the following private key used
to decrypt audit logs securely in a safe location.
Consider the following recommendations:
- Save it to an encrypted storage device or file.
- Use strong, unique passwords to protect the storage device or
file.
- Restrict access to the private key file to authorized
individuals only.

The Private key is:

-----BEGIN PRIVATE KEY-----
<private_key_id>
-----END PRIVATE KEY-----
```

從 **Tiebreaker 1.4** 升級至 **1.6**

請依照下列步驟、將 Tiebreaker 1.4 軟體版本升級至 Tiebreaker 1.6 。

步驟

a. 在執行下列命令 `[root@mcctb ~] #` 提示升級軟體：

```
sh MetroClusterTiebreakerInstall-1.6
```

系統會顯示下列輸出以成功升級：

```
Extracting the MetroCluster Tiebreaker installation/upgrade
archive
Install digest hash is Ok
Performing the MetroCluster Tiebreaker code signature check
Install code signature is Ok
Enter unix user account to use for the installation:
mcctbuseradmin1
Unix user account "mcctbuseradmin1" doesn't exist. Do you wish to
create "mcctbuseradmin1" user account? [Y/N]: y
Unix account "mcctbuseradmin1" created.
Changing password for user mcctbuseradmin1.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.

Enter database user name : root

Please enter database password for root
Enter password:

Password updated successfully in the database.

MetroCluster Tiebreaker requires unix user account
"mcctbuseradmin1" to be added to the group "mcctbgrp" for admin
access.
Do you wish to add ? [Y/N]: y
Unix user account "mcctbuseradmin1" added to "mcctbgrp".
Do you wish to generate your own public-private key pair for
encrypting audit log? [Y/N]: y
Generating public-private key pair...
Configuring Vault...
Starting vault server...
==> Vault server configuration:

             Api Address: <api_addess>
                     Cgo: disabled
```

```
         Cluster Address: <cluster_address>
   Environment Variables: BASH_FUNC_which%%,
DBUS_SESSION_BUS_ADDRESS, GODEBUG, HISTCONTROL, HISTSIZE, HOME,
HOSTNAME, HOST_ACCOUNT, LANG, LESSOPEN, LOGNAME, LS_COLORS, MAIL,
PATH, PWD, SHELL, SHLVL, SSH_CLIENT, SSH_CONNECTION, SSH_TTY,
STAF_TEMP_DIR, TERM, USER, VAULT_ADDR, VAULT_TOKEN,
XDG_RUNTIME_DIR, XDG_SESSION_ID, _, vault_Addr, which_declare
             Go Version: go1.20.5
             Listener 1: tcp (addr: "0.0.0.0:8200", cluster
address: "0.0.0.0:8201", max_request_duration: "1m30s",
max_request_size: "33554432", tls: "enabled")
              Log Level:
                  Mlock: supported: true, enabled: true
          Recovery Mode: false
                Storage: file
                Version: Vault v1.14.0, built 2023-06-
19T11:40:23Z
            Version Sha:
13a649f860186dffe3f3a4459814d87191efc321

==> Vault server started! Log data will stream in below:

2023-11-23T15:58:10.400+0530 [INFO]  proxy environment:
http_proxy="" https_proxy="" no_proxy=""
2023-11-23T15:58:10.432+0530 [INFO]  core: Initializing version
history cache for core
2023-11-23T15:58:20.422+0530 [INFO]  core: security barrier not
initialized
2023-11-23T15:58:20.422+0530 [INFO]  core: seal configuration
missing, not initialized
2023-11-23T15:58:20.424+0530 [INFO]  core: security barrier not
initialized
2023-11-23T15:58:20.425+0530 [INFO]  core: security barrier
initialized: stored=1 shares=5 threshold=3
2023-11-23T15:58:20.427+0530 [INFO]  core: post-unseal setup
starting
2023-11-23T15:58:20.448+0530 [INFO]  core: loaded wrapping token
key
2023-11-23T15:58:20.448+0530 [INFO]  core: successfully setup
plugin catalog: plugin-directory=""
2023-11-23T15:58:20.448+0530 [INFO]  core: no mounts; adding
default mount table
2023-11-23T15:58:20.449+0530 [INFO]  core: successfully mounted:
type=cubbyhole version="v1.14.0+builtin.vault" path=cubbyhole/
namespace="ID: root. Path: "
2023-11-23T15:58:20.449+0530 [INFO]  core: successfully mounted:
```

```
type=system version="v1.14.0+builtin.vault" path=sys/
namespace="ID: root. Path: "
2023-11-23T15:58:20.449+0530 [INFO]  core: successfully mounted:
type=identity version="v1.14.0+builtin.vault" path=identity/
namespace="ID: root. Path: "
2023-11-23T15:58:20.451+0530 [INFO]  core: successfully mounted:
type=token version="v1.14.0+builtin.vault" path=token/
namespace="ID: root. Path: "
2023-11-23T15:58:20.452+0530 [INFO]  rollback: starting rollback
manager
2023-11-23T15:58:20.452+0530 [INFO]  core: restoring leases
2023-11-23T15:58:20.453+0530 [INFO]  identity: entities restored
2023-11-23T15:58:20.453+0530 [INFO]  identity: groups restored
2023-11-23T15:58:20.453+0530 [INFO]  expiration: lease restore
complete
2023-11-23T15:58:20.453+0530 [INFO]  core: usage gauge collection
is disabled
2023-11-23T15:58:20.453+0530 [INFO]  core: Recorded vault
version: vault version=1.14.0 upgrade time="2023-11-23
10:28:20.453481904 +0000 UTC" build date=2023-06-19T11:40:23Z
2023-11-23T15:58:20.818+0530 [INFO]  core: post-unseal setup
complete
2023-11-23T15:58:20.819+0530 [INFO]  core: root token generated
2023-11-23T15:58:20.819+0530 [INFO]  core: pre-seal teardown
starting
2023-11-23T15:58:20.819+0530 [INFO]  rollback: stopping rollback
manager
2023-11-23T15:58:20.819+0530 [INFO]  core: pre-seal teardown
complete
2023-11-23T15:58:21.116+0530 [INFO]  core.cluster-listener.tcp:
starting listener: listener_address=0.0.0.0:8201
2023-11-23T15:58:21.116+0530 [INFO]  core.cluster-listener:
serving cluster requests: cluster_listen_address=[::]:8201
2023-11-23T15:58:21.117+0530 [INFO]  core: post-unseal setup
starting
2023-11-23T15:58:21.117+0530 [INFO]  core: loaded wrapping token
key
2023-11-23T15:58:21.117+0530 [INFO]  core: successfully setup
plugin catalog: plugin-directory=""
2023-11-23T15:58:21.119+0530 [INFO]  core: successfully mounted:
type=system version="v1.14.0+builtin.vault" path=sys/
namespace="ID: root. Path: "
2023-11-23T15:58:21.120+0530 [INFO]  core: successfully mounted:
type=identity version="v1.14.0+builtin.vault" path=identity/
namespace="ID: root. Path: "
2023-11-23T15:58:21.120+0530 [INFO]  core: successfully mounted:
```

```
type=cubbyhole version="v1.14.0+builtin.vault" path=cubbyhole/
namespace="ID: root. Path: "
2023-11-23T15:58:21.123+0530 [INFO]  core: successfully mounted:
type=token version="v1.14.0+builtin.vault" path=token/
namespace="ID: root. Path: "
2023-11-23T15:58:21.123+0530 [INFO]  rollback: starting rollback
manager
2023-11-23T15:58:21.124+0530 [INFO]  core: restoring leases
2023-11-23T15:58:21.124+0530 [INFO]  identity: entities restored
2023-11-23T15:58:21.124+0530 [INFO]  identity: groups restored
2023-11-23T15:58:21.124+0530 [INFO]  expiration: lease restore
complete
2023-11-23T15:58:21.125+0530 [INFO]  core: usage gauge collection
is disabled
2023-11-23T15:58:21.125+0530 [INFO]  core: post-unseal setup
complete
2023-11-23T15:58:21.125+0530 [INFO]  core: vault is unsealed
Success! Uploaded policy: mcctb-policy
2023-11-23T15:58:21.600+0530 [INFO]  core: enabled credential
backend: path=approle/ type=approle version=""
Success! Enabled approle auth method at: approle/
2023-11-23T15:58:21.690+0530 [INFO]  core: successful mount:
namespace="" path=mcctb/ type=kv version=""
Success! Enabled the kv secrets engine at: mcctb/
Success! Data written to: auth/approle/role/mcctb-app
Upgrading to NetApp-MetroCluster-Tiebreaker-Software-1.6-
1.x86_64.rpm
Preparing...
################################# [100%]
Updating / installing...
   1:NetApp-MetroCluster-Tiebreaker-
So################################# [ 50%]
Performing file integrity check
etc/cron.weekly/metrocluster-tiebreaker-support is Ok
etc/cron.weekly/metrocluster-tiebreaker-support-cov is Ok
etc/init.d/netapp-metrocluster-tiebreaker-software is Ok
etc/init.d/netapp-metrocluster-tiebreaker-software-cov is Ok
etc/logrotate.d/mcctb is Ok
opt/netapp/mcctb/lib/common/activation-1.1.1.jar is Ok
opt/netapp/mcctb/lib/common/aopalliance.jar is Ok
opt/netapp/mcctb/lib/common/args4j.jar is Ok
opt/netapp/mcctb/lib/common/aspectjrt.jar is Ok
opt/netapp/mcctb/lib/common/aspectjweaver.jar is Ok
opt/netapp/mcctb/lib/common/asup.jar is Ok
opt/netapp/mcctb/lib/common/bcpkix-jdk15on.jar is Ok
opt/netapp/mcctb/lib/common/bcprov-jdk15on.jar is Ok
```

```
opt/netapp/mcctb/lib/common/bcprov-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/bctls-fips-1.0.13.jar is Ok
opt/netapp/mcctb/lib/common/bctls-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/bcutil-jdk18on.jar is Ok
opt/netapp/mcctb/lib/common/cglib.jar is Ok
opt/netapp/mcctb/lib/common/commons-codec.jar is Ok
opt/netapp/mcctb/lib/common/commons-collections4.jar is Ok
opt/netapp/mcctb/lib/common/commons-compress.jar is Ok
opt/netapp/mcctb/lib/common/commons-daemon.jar is Ok
opt/netapp/mcctb/lib/common/commons-daemon.src.jar is Ok
opt/netapp/mcctb/lib/common/commons-dbcp2.jar is Ok
opt/netapp/mcctb/lib/common/commons-io.jar is Ok
opt/netapp/mcctb/lib/common/commons-lang3.jar is Ok
opt/netapp/mcctb/lib/common/commons-logging.jar is Ok
opt/netapp/mcctb/lib/common/commons-pool2.jar is Ok
opt/netapp/mcctb/lib/common/guava.jar is Ok
opt/netapp/mcctb/lib/common/httpclient.jar is Ok
opt/netapp/mcctb/lib/common/httpcore.jar is Ok
opt/netapp/mcctb/lib/common/jakarta.activation.jar is Ok
opt/netapp/mcctb/lib/common/jakarta.xml.bind-api.jar is Ok
opt/netapp/mcctb/lib/common/java-xmlbuilder.jar is Ok
opt/netapp/mcctb/lib/common/javax.inject.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-api-2.3.1.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-core.jar is Ok
opt/netapp/mcctb/lib/common/jaxb-impl.jar is Ok
opt/netapp/mcctb/lib/common/jline.jar is Ok
opt/netapp/mcctb/lib/common/jna.jar is Ok
opt/netapp/mcctb/lib/common/joda-time.jar is Ok
opt/netapp/mcctb/lib/common/jsch.jar is Ok
opt/netapp/mcctb/lib/common/json.jar is Ok
opt/netapp/mcctb/lib/common/jsvc.zip is Ok
opt/netapp/mcctb/lib/common/junixsocket-common.jar is Ok
opt/netapp/mcctb/lib/common/junixsocket-native-common.jar is Ok
opt/netapp/mcctb/lib/common/logback-classic.jar is Ok
opt/netapp/mcctb/lib/common/logback-core.jar is Ok
opt/netapp/mcctb/lib/common/mail-1.6.2.jar is Ok
opt/netapp/mcctb/lib/common/mariadb-java-client.jar is Ok
opt/netapp/mcctb/lib/common/mcctb-mib.jar is Ok
opt/netapp/mcctb/lib/common/mcctb.jar is Ok
opt/netapp/mcctb/lib/common/mockito-core.jar is Ok
opt/netapp/mcctb/lib/common/slf4j-api.jar is Ok
opt/netapp/mcctb/lib/common/snmp4j.jar is Ok
opt/netapp/mcctb/lib/common/spring-aop.jar is Ok
opt/netapp/mcctb/lib/common/spring-beans.jar is Ok
opt/netapp/mcctb/lib/common/spring-context-support.jar is Ok
opt/netapp/mcctb/lib/common/spring-context.jar is Ok
```

```
opt/netapp/mcctb/lib/common/spring-core.jar is Ok
opt/netapp/mcctb/lib/common/spring-expression.jar is Ok
opt/netapp/mcctb/lib/common/spring-web.jar is Ok
opt/netapp/mcctb/lib/common/vault-java-driver.jar is Ok
opt/netapp/mcctb/lib/common/xz.jar is Ok
opt/netapp/mcctb/lib/org.jacoco.agent-0.8.8-runtime.jar is Ok
opt/netapp/mcctb/bin/mcctb-asup-invoke is Ok
opt/netapp/mcctb/bin/mcctb_postrotate is Ok
opt/netapp/mcctb/bin/netapp-metrocluster-tiebreaker-software-cli
is Ok
/

Synchronizing state of netapp-metrocluster-tiebreaker-
software.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable netapp-
metrocluster-tiebreaker-software

Attempting to start NetApp MetroCluster Tiebreaker software
services
Started NetApp MetroCluster Tiebreaker software services
Successfully upgraded NetApp MetroCluster Tiebreaker software to
version 1.6.
Cleaning up / removing...
   2:NetApp-MetroCluster-Tiebreaker-
So################################ [100%]


 IMPORTANT: Please ensure you store the following private key
used to decrypt audit logs securely in a safe location.
Consider the following recommendations:
- Save it to an encrypted storage device or file.
- Use strong, unique passwords to protect the storage device or
file.
- Restrict access to the private key file to authorized
individuals only.

The Private key is:

-----BEGIN PRIVATE KEY-----
<private_key_id>
-----END PRIVATE KEY-----
```

安裝斷路器 1.5 或 1.4

設定管理員對ONTAP SURE API和SSH的存取權限

您可以設定管理員存取ONTAP 功能以存取SURE API和SSH。

步驟

1. 建立可ONTAP 存取API的管理員使用者： `security login create -user-or-group-name` *`mcctb`* `-application ontapi -authentication-method` *`password`*
2. 建立具有SSH存取權限的管理使用者： `security login create -user-or-group-name` *`mcctb`* `-application` *`ssh`* `-authentication-method` *`password`*
3. 確認已建立新的管理使用者： `security login show`
4. 在合作夥伴叢集上重複這些步驟。

> "系統管理員驗證與RBAC" 已實作。

安裝MetroCluster 相依的斷點器

視主機 Linux 作業系統而定、您必須先安裝 MySQL 或 MariaDB 伺服器、才能安裝或升級 tiebreaker 軟體。

步驟

1. 安裝JRE.

   安裝JRE

2. 安裝及設定Vault。

   安裝及設定Vault

3. 安裝MySQL或MariaDB伺服器：

如果Linux主機是	然後…
Red Hat Enterprise Linux 7/CentOS 7.	安裝MySQL 在Red Hat Enterprise Linux 7或CentOS 7上安裝MySQL Server 5.5.30或更新版本及5.6.x版本
Red Hat Enterprise Linux 8	安裝MariaDB 在Red Hat Enterprise Linux 8上安裝MariaDB伺服器

安裝JRE

安裝或升級Tiebreaker軟體之前、您必須先在主機系統上安裝JRE。對於執行Tiebreaker 1.4及更早版本的系統、請執行JRE 8。對於執行tiebreaker 1.5及更新版本的系統、請執行OpenJDK 17、18或19。

> 以下範例中的輸出顯示了 JRE 1.8.0 。（JRE 8）。

步驟

1. 以「root」使用者或可變更為進階權限模式的Sudo使用者身分登入。

```
login as: root
root@mcctb's password:
Last login: Fri Jan  8 21:33:00 2017 from host.domain.com
```

2. 檢查可用的 JRE 版本：

```
yum search openjdk
```

3. 針對您要安裝的 tiebreaker 版本安裝適當的 JRE 版本：

```
yum install java-<version>-openjdk.x86_64
```

```
[root@mcctb ~]# yum install java-1.8.0-openjdk.x86_64
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
... shortened....
Dependencies Resolved

=========================================================================
Package                Arch   Version                  Repository    Size
=========================================================================
Installing:
 java-1.8.0-openjdk  x86_64  1:1.8.0.144-0.b01.el7_4 updates      238 k
 ..
 ..
Transaction Summary
=========================================================================
Install  1 Package  (+ 4 Dependent packages)

Total download size: 34 M
Is this ok [y/d/N]: y

Installed:
java-1.8.0-openjdk.x86_64 1:1.8.0.144-0.b01.el7_4
Complete!
```

安裝及設定Vault

如果您沒有或想要使用本機Vault伺服器、則必須安裝Vault。您可以參閱本標準程序來安裝Vault、或參閱《橋式安裝說明》以取得替代準則。

> 如果您的網路中有Vault伺服器、您可以設定MetroCluster 使用該Vault安裝的還原斷路器主機。如果您這麼做、就不需要在主機上安裝 Vault 。

步驟

1. 瀏覽至 `/bin` 目錄：

```
[root@mcctb] cd /bin
```

2. 下載Vault壓縮檔。

```
[root@mcctb /bin]#  curl -sO
https://releases.hashicorp.com/vault/1.12.2/vault_1.12.2_linux_amd64.zip
```

3. 解壓縮Vault檔案。

```
[root@mcctb /bin]# unzip vault_1.12.2_linux_amd64.zip
Archive:  vault_1.12.2_linux_amd64.zip
  inflating: vault
```

4. 驗證安裝。

```
[root@mcctb /bin]# vault -version
Vault v1.12.2 (415e1fe3118eebd5df6cb60d13defdc01aa17b03), built 2022-11-
23T12:53:46Z
```

5. 瀏覽至 `/root` 目錄：

```
[root@mcctb /bin] cd /root
```

6. 在下建立 Vault 組態檔案 `/root` 目錄。

   的 `[root@mcctb ~]` 提示、複製並執行下列命令以建立 `config.hcl` 檔案：

```
# cat > config.hcl << EOF
 storage "file" {
  address = "127.0.0.1:8500"
  path    = "/mcctb_vdata/data"
 }
 listener "tcp" {
   address     = "127.0.0.1:8200"
   tls_disable = 1
 }
EOF
```

7. 啟動Vault伺服器：

```
[root@mcctb ~] vault server -config config.hcl &
```

8. 匯出Vault位址。

```
[root@mcctb ~]# export VAULT_ADDR="http://127.0.0.1:8200"
```

9. 初始化Vault。

```
[root@mcctb ~]# vault operator init
2022-12-15T14:57:22.113+0530 [INFO]  core: security barrier not
initialized
2022-12-15T14:57:22.113+0530 [INFO]  core: seal configuration missing,
not initialized
2022-12-15T14:57:22.114+0530 [INFO]  core: security barrier not
initialized
2022-12-15T14:57:22.116+0530 [INFO]  core: security barrier initialized:
stored=1 shares=5 threshold=3
2022-12-15T14:57:22.118+0530 [INFO]  core: post-unseal setup starting
2022-12-15T14:57:22.137+0530 [INFO]  core: loaded wrapping token key
2022-12-15T14:57:22.137+0530 [INFO]  core: Recorded vault version: vault
version=1.12.2 upgrade time="2022-12-15 09:27:22.137200412 +0000 UTC"
build date=2022-11-23T12:53:46Z
2022-12-15T14:57:22.137+0530 [INFO]  core: successfully setup plugin
catalog: plugin-directory=""
2022-12-15T14:57:22.137+0530 [INFO]  core: no mounts; adding default
mount table
2022-12-15T14:57:22.143+0530 [INFO]  core: successfully mounted backend:
type=cubbyhole version="" path=cubbyhole/
2022-12-15T14:57:22.144+0530 [INFO]  core: successfully mounted backend:
```

```
type=system version="" path=sys/
2022-12-15T14:57:22.144+0530 [INFO]  core: successfully mounted backend:
type=identity version="" path=identity/
2022-12-15T14:57:22.148+0530 [INFO]  core: successfully enabled
credential backend: type=token version="" path=token/ namespace="ID:
root. Path: "
2022-12-15T14:57:22.149+0530 [INFO]  rollback: starting rollback manager
2022-12-15T14:57:22.149+0530 [INFO]  core: restoring leases
2022-12-15T14:57:22.150+0530 [INFO]  expiration: lease restore complete
2022-12-15T14:57:22.150+0530 [INFO]  identity: entities restored
2022-12-15T14:57:22.150+0530 [INFO]  identity: groups restored
2022-12-15T14:57:22.151+0530 [INFO]  core: usage gauge collection is
disabled
2022-12-15T14:57:23.385+0530 [INFO]  core: post-unseal setup complete
2022-12-15T14:57:23.387+0530 [INFO]  core: root token generated
2022-12-15T14:57:23.387+0530 [INFO]  core: pre-seal teardown starting
2022-12-15T14:57:23.387+0530 [INFO]  rollback: stopping rollback manager
2022-12-15T14:57:23.387+0530 [INFO]  core: pre-seal teardown complete
Unseal Key 1: <unseal_key_1_id>
Unseal Key 2: <unseal_key_2_id>
Unseal Key 3: <unseal_key_3_id>
Unseal Key 4: <unseal_key_4_id>
Unseal Key 5: <unseal_key_5_id>

Initial Root Token: <initial_root_token_id>


Vault initialized with 5 key shares and a key threshold of 3. Please
securely
distribute the key shares printed above. When the Vault is re-sealed,
restarted, or stopped, you must supply at least 3 of these keys to
unseal it
before it can start servicing requests.

Vault does not store the generated root key. Without at least 3 keys to
reconstruct the root key, Vault will remain permanently sealed!

It is possible to generate new unseal keys, provided you have a quorum
of
existing unseal keys shares. See "vault operator rekey" for more
information.
```

您必須將金鑰 ID 和初始根權杖記錄並儲存在安全位置、以便稍後在程序中使用。

10. 匯出Vault根憑證。

```
[root@mcctb ~]#  export VAULT_TOKEN="<initial_root_token_id>"
```

11. 使用建立的五個金鑰中的任何三個來解除保存 Vault 的密封。

    您必須執行 `vault operator unseal` 三個按鍵的命令：

    a. 使用第一個金鑰來解除保存資料保險箱的密封：

    ```
    [root@mcctb ~]# vault operator unseal
    Unseal Key (will be hidden):
    Key                Value
    ---                -----
    Seal Type          shamir
    Initialized        true
    Sealed             true
    Total Shares       5
    Threshold          3
    Unseal Progress    1/3
    Unseal Nonce       <unseal_key_1_id>
    Version            1.12.2
    Build Date         2022-11-23T12:53:46Z
    Storage Type       file
    HA Enabled         false
    ```

    b. 使用第二個金鑰來解除保存資料保險箱的密封：

    ```
    [root@mcctb ~]# vault operator unseal
    Unseal Key (will be hidden):
    Key                Value
    ---                -----
    Seal Type          shamir
    Initialized        true
    Sealed             true
    Total Shares       5
    Threshold          3
    Unseal Progress    2/3
    Unseal Nonce       <unseal_key_2_id>
    Version            1.12.2
    Build Date         2022-11-23T12:53:46Z
    Storage Type       file
    HA Enabled         false
    ```

    c. 使用第三個金鑰來解除保存資料保險箱的密封：

```
[root@mcctb ~]# vault operator unseal
Unseal Key (will be hidden):
2022-12-15T15:15:00.980+0530 [INFO]  core.cluster-listener.tcp:
starting listener: listener_address=127.0.0.1:8201
2022-12-15T15:15:00.980+0530 [INFO]  core.cluster-listener: serving
cluster requests: cluster_listen_address=127.0.0.1:8201
2022-12-15T15:15:00.981+0530 [INFO]  core: post-unseal setup starting
2022-12-15T15:15:00.981+0530 [INFO]  core: loaded wrapping token key
2022-12-15T15:15:00.982+0530 [INFO]  core: successfully setup plugin
catalog: plugin-directory=""
2022-12-15T15:15:00.983+0530 [INFO]  core: successfully mounted
backend: type=system version="" path=sys/
2022-12-15T15:15:00.984+0530 [INFO]  core: successfully mounted
backend: type=identity version="" path=identity/
2022-12-15T15:15:00.984+0530 [INFO]  core: successfully mounted
backend: type=cubbyhole version="" path=cubbyhole/
2022-12-15T15:15:00.986+0530 [INFO]  core: successfully enabled
credential backend: type=token version="" path=token/ namespace="ID:
root. Path: "
2022-12-15T15:15:00.986+0530 [INFO]  rollback: starting rollback
manager
2022-12-15T15:15:00.987+0530 [INFO]  core: restoring leases
2022-12-15T15:15:00.987+0530 [INFO]  expiration: lease restore
complete
2022-12-15T15:15:00.987+0530 [INFO]  identity: entities restored
2022-12-15T15:15:00.987+0530 [INFO]  identity: groups restored
2022-12-15T15:15:00.988+0530 [INFO]  core: usage gauge collection is
disabled
2022-12-15T15:15:00.989+0530 [INFO]  core: post-unseal setup complete
2022-12-15T15:15:00.989+0530 [INFO]  core: vault is unsealed
Key             Value
---             -----
Seal Type       shamir
Initialized     true
Sealed          false
Total Shares    5
Threshold       3
Version         1.12.2
Build Date      2022-11-23T12:53:46Z
Storage Type    file
Cluster Name    vault-cluster
Cluster ID      <cluster_id>
HA Enabled      false
```

12. 確認Vault密封狀態為假。

```
[root@mcctb ~]# vault status
Key             Value
---             -----
Seal Type       shamir
Initialized     true
Sealed          false
Total Shares    5
Threshold       3
Version         1.12.2
Build Date      2022-11-23T12:53:46Z
Storage Type    file
Cluster Name    vault-cluster
Cluster ID      <cluster_id>
HA Enabled      false
```

13. 將 Vault 服務設定為開機時啟動。
    a. 執行下列命令： `cd /etc/systemd/system`

    ```
    [root@mcctb ~]#  cd /etc/systemd/system
    ```

    b. 的 `[root@mcctb system]` 提示、複製並執行下列命令以建立 Vault 服務檔案。

    ```
    # cat > vault.service << EOF
    [Unit]
    Description=Vault Service
    After=mariadb.service

    [Service]
    Type=forking
    ExecStart=/usr/bin/vault server -config /root/config.hcl &
    Restart=on-failure

    [Install]
    WantedBy=multi-user.target
    EOF
    ```

    c. 執行下列命令： `systemctl daemon-reload`

    ```
    [root@mcctb system]#  systemctl daemon-reload
    ```

    d. 執行下列命令： `systemctl enable vault.service`

```
[root@mcctb system]#  systemctl enable vault.service
Created symlink /etc/systemd/system/multi-
user.target.wants/vault.service → /etc/systemd/system/vault.service.
```

安裝MetroCluster 完ESITiebreaker時、系統會提示您使用此功能。如果您想要變更解封Vault的方法、則需要解除安裝並重新安裝MetroCluster 還原器軟體。

在**Red Hat Enterprise Linux 7**或**CentOS 7**上安裝**MySQL Server 5.5.30**或更新版本及**5.6.x**版本

安裝或升級Tiebreaker軟體之前、您必須先在主機系統上安裝MySQL Server 5.5.30或更新版本及5.6.x版本。

步驟

1. 以root使用者或可變更為進階權限模式的Sudo使用者身分登入。

```
login as: root
root@mcctb's password:
Last login: Fri Jan  8 21:33:00 2016 from host.domain.com
```

2. 將MySQL儲存庫新增至主機系統：

   [root@mcctb ~]# yum localinstall https://dev.mysql.com/get/mysql57-community-release-el6-11.noarch.rpm`

```
Loaded plugins: product-id, refresh-packagekit, security, subscription-
manager
Setting up Local Package Process
Examining /var/tmp/yum-root-LLUw0r/mysql-community-release-el6-
5.noarch.rpm: mysql-community-release-el6-5.noarch
Marking /var/tmp/yum-root-LLUw0r/mysql-community-release-el6-
5.noarch.rpm to be installed
Resolving Dependencies
--> Running transaction check
---> Package mysql-community-release.noarch 0:el6-5 will be installed
--> Finished Dependency Resolution
Dependencies Resolved
=============================================================================
========
Package               Arch   Version
                                     Repository
Size
=============================================================================
========
Installing:
mysql-community-release
                       noarch el6-5 /mysql-community-release-el6-
5.noarch 4.3 k
Transaction Summary
=============================================================================
========
Install       1 Package(s)
Total size: 4.3 k
Installed size: 4.3 k
Is this ok [y/N]: y
Downloading Packages:
Running rpm_check_debug
Running Transaction Test
Transaction Test Succeeded
Running Transaction
  Installing : mysql-community-release-el6-5.noarch
1/1
  Verifying  : mysql-community-release-el6-5.noarch
1/1
Installed:
  mysql-community-release.noarch 0:el6-5
Complete!
```

3. 停用MySQL 57儲存庫：

[root@mcctb ~]# yum-config-manager -disablemysql57-community

4. 啟用MySQL 56儲存庫：

   [root@mcctb ~]# yum-config-manager --enable mysql56-community

5. 啟用儲存庫：

   [root@mcctb ~]# yum repolist enabled| grep "mysql.**-community.**

```
mysql-connectors-community           MySQL Connectors Community
21
mysql-tools-community                MySQL Tools Community
35
mysql56-community                    MySQL 5.6 Community Server
231
```

6. 安裝MySQL社群伺服器：

   [root@mcctb ~]# yum install mysql-community-server'

```
Loaded plugins: product-id, refresh-packagekit, security, subscription-
manager
This system is not registered to Red Hat Subscription Management. You
can use subscription-manager
to register.
Setting up Install Process
Resolving Dependencies
--> Running transaction check
 .....Output truncated.....
---> Package mysql-community-libs-compat.x86_64 0:5.6.29-2.el6 will be
obsoleting
--> Finished Dependency Resolution
Dependencies Resolved
================================================================================
======
Package                          Arch   Version       Repository
Size
================================================================================
======
Installing:
 mysql-community-client         x86_64  5.6.29-2.el6  mysql56-community
18  M
     replacing  mysql.x86_64 5.1.71-1.el6
 mysql-community-libs           x86_64  5.6.29-2.el6  mysql56-community
1.9 M
     replacing  mysql-libs.x86_64 5.1.71-1.el6
```

```
 mysql-community-libs-compat    x86_64  5.6.29-2.el6  mysql56-community
1.6 M
     replacing  mysql-libs.x86_64 5.1.71-1.el6
 mysql-community-server         x86_64  5.6.29-2.el6  mysql56-community
53  M
     replacing  mysql-server.x86_64 5.1.71-1.el6
Installing for dependencies:
mysql-community-common          x86_64  5.6.29-2.el6  mysql56-community
308 k

Transaction Summary
================================================================================
=======
Install       5 Package(s)
Total download size: 74 M
Is this ok [y/N]: y
Downloading Packages:
(1/5): mysql-community-client-5.6.29-2.el6.x86_64.rpm       |  18 MB
00:28
(2/5): mysql-community-common-5.6.29-2.el6.x86_64.rpm       | 308 kB
00:01
(3/5): mysql-community-libs-5.6.29-2.el6.x86_64.rpm         | 1.9 MB
00:05
(4/5): mysql-community-libs-compat-5.6.29-2.el6.x86_64.rpm  | 1.6 MB
00:05
(5/5): mysql-community-server-5.6.29-2.el6.x86_64.rpm       |  53 MB
03:42
 --------------------------------------------------------------------------
--------
Total                                                 289 kB/s |  74 MB
04:24
warning: rpmts_HdrFromFdno: Header V3 DSA/SHA1 Signature, key ID
<key_id> NOKEY
Retrieving key from file:/etc/pki/rpm-gpg/RPM-GPG-KEY-mysql
Importing GPG key 0x5072E1F5:
 Userid : MySQL Release Engineering <mysql-build@oss.oracle.com>
Package: mysql-community-release-el6-5.noarch
         (@/mysql-community-release-el6-5.noarch)
 From   : file:/etc/pki/rpm-gpg/RPM-GPG-KEY-mysql
Is this ok [y/N]: y
Running rpm_check_debug
Running Transaction Test
Transaction Test Succeeded
Running Transaction
  Installing : mysql-community-common-5.6.29-2.el6.x86_64
....Output truncated....
```

```
1.el6.x86_64
7/8
  Verifying  : mysql-5.1.71-1.el6.x86_64
8/8
Installed:
  mysql-community-client.x86_64 0:5.6.29-2.el6
  mysql-community-libs.x86_64 0:5.6.29-2.el6
  mysql-community-libs-compat.x86_64 0:5.6.29-2.el6
  mysql-community-server.x86_64 0:5.6.29-2.el6

Dependency Installed:
  mysql-community-common.x86_64 0:5.6.29-2.el6

Replaced:
  mysql.x86_64 0:5.1.71-1.el6 mysql-libs.x86_64 0:5.1.71-1.el6
  mysql-server.x86_64 0:5.1.71-1.el6
Complete!
```

7. 啟動MySQL伺服器：

```
Initializing MySQL database:  2016-04-05 19:44:38 0 [Warning] TIMESTAMP
with implicit DEFAULT value is deprecated. Please use
--explicit_defaults_for_timestamp server option (see documentation
for more details).
2016-04-05 19:44:38 0 [Note] /usr/sbin/mysqld (mysqld 5.6.29)
        starting as process 2487 ...
2016-04-05 19:44:38 2487 [Note] InnoDB: Using atomics to ref count
        buffer pool pages
2016-04-05 19:44:38 2487 [Note] InnoDB: The InnoDB memory heap is
disabled
....Output truncated....
2016-04-05 19:44:42 2509 [Note] InnoDB: Shutdown completed; log sequence
       number 1625987

PLEASE REMEMBER TO SET A PASSWORD FOR THE MySQL root USER!
To do so, start the server, then issue the following commands:

  /usr/bin/mysqladmin -u root password 'new-password'
  /usr/bin/mysqladmin -u root -h mcctb password 'new-password'

Alternatively, you can run:
  /usr/bin/mysql_secure_installation

which will also give you the option of removing the test
databases and anonymous user created by default.  This is
strongly recommended for production servers.
.....Output truncated.....
WARNING: Default config file /etc/my.cnf exists on the system
This file will be read by default by the MySQL server
If you do not want to use this, either remove it, or use the
--defaults-file argument to mysqld_safe when starting the server

                                                           [  OK  ]
Starting mysqld:                                           [  OK  ]
```

8. 確認MySQL伺服器正在執行：

   [root@mcctb ~]#服務mysqLD狀態

```
mysqld (pid  2739) is running...
```

9. 設定安全性和密碼設定：

```
NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MySQL
```

```
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

 In order to log into MySQL to secure it, we'll need the current
 password for the root user.  If you've just installed MySQL, and
 you haven't set the root password yet, the password will be blank,
 so you should just press enter here.

 Enter current password for root (enter for none):   <== on default
install
                                                         hit enter here
 OK, successfully used password, moving on...

 Setting the root password ensures that nobody can log into the MySQL
 root user without the proper authorization.

 Set root password? [Y/n] y
 New password:
 Re-enter new password:
 Password updated successfully!
 Reloading privilege tables..
  ... Success!

 By default, a MySQL installation has an anonymous user, allowing anyone
 to log into MySQL without having to have a user account created for
 them.  This is intended only for testing, and to make the installation
 go a bit smoother.  You should remove them before moving into a
 production environment.

 Remove anonymous users? [Y/n] y
  ... Success!

 Normally, root should only be allowed to connect from 'localhost'.
This
 ensures that someone cannot guess at the root password from the
network.

 Disallow root login remotely? [Y/n] y
  ... Success!

 By default, MySQL comes with a database named 'test' that anyone can
 access.  This is also intended only for testing, and should be removed
 before moving into a production environment.

 Remove test database and access to it? [Y/n] y
  - Dropping test database...
 ERROR 1008 (HY000) at line 1: Can't drop database 'test';
 database doesn't exist
```

```
 ... Failed!  Not critical, keep moving...
 - Removing privileges on test database...
 ... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
 ... Success!

All done!  If you've completed all of the above steps, your MySQL
installation should now be secure.

Thanks for using MySQL!

Cleaning up...
```

10. 驗證MySQL登入是否正常運作：

```
Enter password: <configured_password>
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 5.6.29 MySQL Community Server (GPL)

Copyright (c) 2000, 2016, Oracle and/or its affiliates. All rights
reserved.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input
statement.
mysql>
```

如果MySQL登入正常運作、輸出將在「mysql>」提示字元結束。

啟用**MySQL**自動啟動設定

您應該確認MySQL精靈的自動啟動功能已開啟。開啟MySQL精靈會在MetroCluster 系統重新開機時、自動重新啟動MySQL。如果MySQL精靈未執行、則tiebreaker軟體會繼續執行、但無法重新啟動、也無法進行組態變更。

步驟

1. 驗證MySQL在開機時是否已啟用自動啟動：

[root@mcctb ~]# systemctl list-unit-filesmysqld.service`

```
UNIT FILE          State
------------------ ----------
mysqld.service     enabled
```

如果MySQL在開機時未啟用自動啟動、請參閱MySQL文件、以啟用安裝的自動啟動功能。

在Red Hat Enterprise Linux 8上安裝MariaDB伺服器

您必須先在主機系統上安裝MariaDB伺服器、才能安裝或升級tiebreaker軟體。

開始之前

您的主機系統必須在Red Hat Enterprise Linux（RHEL）8上執行。

步驟

1. 以登入 `root` 使用者或可Sudo進階權限模式的使用者。

```
login as: root
root@mcctb's password:
Last login: Fri Jan  8 21:33:00 2017 from host.domain.com
```

1. 安裝 MariaDB 伺服器：

   [root@mcctb ~]# yum install MariaDB-server.x86_64

```
[root@mcctb ~]# yum install mariadb-server.x86_64
Loaded plugins: fastestmirror, langpacks
 ...
 ...

=================================================================
===
 Package                      Arch   Version         Repository
Size
=================================================================
===
Installing:
mariadb-server               x86_64   1:5.5.56-2.el7   base
11 M
Installing for dependencies:

Transaction Summary
=================================================================
```

```
===
Install  1 Package  (+8 Dependent packages)
Upgrade             ( 1 Dependent package)

Total download size: 22 M
Is this ok [y/d/N]: y

Downloading packages:
No Presto metadata available for base warning:
/var/cache/yum/x86_64/7/base/packages/mariadb-libs-5.5.56-
2.el7.x86_64.rpm:
Header V3 RSA/SHA256 Signature,
key ID f4a80eb5: NOKEY] 1.4 MB/s | 3.3 MB  00:00:13 ETA
Public key for mariadb-libs-5.5.56-2.el7.x86_64.rpm is not installed
(1/10): mariadb-libs-5.5.56-2.el7.x86_64.rpm  | 757 kB  00:00:01
..
..
(10/10): perl-Net-Daemon-0.48-5.el7.noarch.rpm|  51 kB  00:00:01
--------------------------------------------------------------------------
-----------------
Installed:
  mariadb-server.x86_64 1:5.5.56-2.el7

Dependency Installed:
mariadb.x86_64 1:5.5.56-2.el7
perl-Compress-Raw-Bzip2.x86_64 0:2.061-3.el7
perl-Compress-Raw-Zlib.x86_64 1:2.061-4.el7
perl-DBD-MySQL.x86_64 0:4.023-5.el7
perl-DBI.x86_64 0:1.627-4.el7
perl-IO-Compress.noarch 0:2.061-2.el7
perl-Net-Daemon.noarch 0:0.48-5.el7
perl-PlRPC.noarch 0:0.2020-14.el7

Dependency Updated:
  mariadb-libs.x86_64 1:5.5.56-2.el7
Complete!
```

2. 啟動MariaDB伺服器：

   [root@mcctb ~]# systemcl start MariaDB

3. 確認MariaDB伺服器已啟動：

   [root@mcctb ~]# systemctl狀態MariaDB

```
[root@mcctb ~]# systemctl status mariadb
mariadb.service - MariaDB database server
...
Nov 08 21:28:59 mcctb systemd[1]: Starting MariaDB database server...
...
Nov 08 21:29:01 mcctb systemd[1]: Started MariaDB database server.
```

4. 設定安全性和密碼設定：

> 當系統提示您輸入 root 密碼時、請將其保留空白、然後按 Enter 鍵繼續設定安全性和密碼設定。

```
root@localhost systemd]# mysql_secure_installation

 NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user.  If you've just installed MariaDB, and
you haven't set the root password yet, the password will be blank,
so you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB
root user without the proper authorisation.

Set root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
 ... Success!


By default, a MariaDB installation has an anonymous user, allowing
anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
 ... Success!
```

```
Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
 ... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access.  This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
 - Dropping test database...
  ... Success!
 - Removing privileges on test database...
   ... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n]

 ... Success!

Cleaning up...

All done!  If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
```

啟用 **MariaDB** 伺服器的自動啟動設定

您應該確認 MariaDB 伺服器的自動啟動功能已開啟。如果您未啟用自動啟動功能、MetroCluster 且駐留的系統必須重新開機、則tiebreaker軟體會繼續執行、但無法重新啟動MariaDB服務、也無法進行組態變更。

步驟

1. 啟用自動啟動服務：

   [root@mcctb ~]# systemctl enable mariadb.service`

2. 確認已啟用MariaDB、以便在開機時自動啟動：

   [root@mcctb ~]# systemctl list-unit-filesmariadb.service`

```
UNIT FILE           State
------------------ ----------
mariadb.service     enabled
```

安裝或升級軟體套件

您必須在MetroCluster 本機電腦上安裝或升級ESITirepreaker軟體、才能監控MetroCluster 各種組態。

關於這項工作

- 您的儲存系統必須執行ONTAP 不含更新版本的資訊。
- 您必須使用安裝OpenJDK `yum install java-x.x.x-openjdk` 命令。對於執行Tiebreaker 1.4及更早版本的系統、請執行JRE 8。對於執行tiebreaker 1.5及更新版本的系統、請執行OpenJDK 17、18或19。範例中的輸出顯示jre 1.8.0。（JRE 8）。
- 您可以將 MetroCluster tiebreaker 安裝為非 root 使用者、並擁有足夠的管理權限來執行 tiebreaker 安裝、建立表格和使用者、以及設定使用者密碼。

步驟

1. 下載 MetroCluster Tiebreaker 軟體。此範例使用1.5版。

   "NetApp 支援網站"

2. 下載 `MetroCluster_Tiebreaker_RPM_GPG` 金鑰：

   "NetApp 支援網站"

3. 以root使用者身分登入主機。
4. 建立非 root 使用者和 `mcctbgrp` 群組：

   a. 建立非 root 使用者並設定密碼。

      下列命令範例可建立名為的非 root 使用者 `mcctbuser1`：

      ```
      [root@mcctb ~]# useradd mcctbuser1
      [root@mcctb ~]# passwd mcctbuser1
      Changing password for user mcctbuser1.
      New password:
      Retype new password:
      passwd: all authentication tokens updated successfully.
      ```

   b. 建立名為的群組 `mcctbgrp`：

      ```
      [root@mcctb ~~]# groupadd mcctbgrp
      ```

   c. 將您建立的非 root 使用者新增至 `mcctbgrp` 群組：

下列命令會新增 `mcctbuser1` 至 `mcctbgrp` 群組：

```
[root@mcctb ~]# usermod -a -G mcctbgrp mcctbuser1
```

5. 驗證RPM檔案。

從包含 RPM 金鑰的目錄執行下列子步驟。

a. 下載並匯入RPM金鑰檔：

```
[root@mcctb ~]# rpm --import MetroCluster_Tiebreaker_RPM_GPG.key
```

b. 檢查指紋、確認已匯入正確的金鑰。

下列範例顯示正確的金鑰指紋：

```
root@mcctb:~/signing/mcctb-rpms# gpg --show-keys --with-fingerprint
MetroCluster_Tiebreaker_RPM_GPG.key
pub   rsa3072 2022-11-17 [SCEA] [expires: 2025-11-16]
      65AC 1562 E28A 1497 7BBD  7251 2855 EB02 3E77 FAE5
uid                      MCCTB-RPM (mcctb RPM production signing)
<mcctb-rpm@netapp.com>
```

a. 驗證簽名： `rpm --checksig NetApp-MetroCluster-Tiebreaker-Software-1.5-1.x86_64.rpm`

```
NetApp-MetroCluster-Tiebreaker-Software-1.5-1.x86_64.rpm: digests OK
```

> 您必須在成功驗證簽名之後、才能繼續安裝。

6. [[install-tiebreaker）]安裝或升級Tiebreaker軟體：

> 從Tiebreaker 1.4版升級時、您只能升級至Tiebreaker 1.5版。不支援從舊版升級至Tiebreaker 1.5。

根據您要執行新安裝或升級現有安裝、從下列選項中選取正確的程序。

執行新安裝

a. 擷取並記錄 Java 的絕對路徑：

```
[root@mcctb ~]# readlink -f /usr/bin/java
/usr/lib/jvm/java-19-openjdk-19.0.0.0.36-
2.rolling.el8.x86_64/bin/java
```

b. 執行下列命令： `rpm -ivh NetApp-MetroCluster-Tiebreaker-Software-1.5-1.x86_64.rpm`

系統會顯示下列輸出、以利成功安裝：

> 在安裝期間出現提示時、請提供您先前建立並指派給的非 root 使用者 `mcctbgrp` 群組：

```
Verifying...
################################# [100%]
Preparing...
################################# [100%]
Updating / installing...
   1:NetApp-MetroCluster-Tiebreaker-
So################################# [100%]
Enter the absolute path for Java : /usr/lib/jvm/java-19-openjdk-
19.0.0.0.36-2.rolling.el8.x86_64/bin/java
Verifying if Java exists...
Found Java. Proceeding with the installation.
Enter host user account to use for the installation:
mcctbuser1
User account mcctbuser1 found. Proceeding with the installation
Enter database user name:
root
Please enter database password for root
Enter password:
Sealed          false
Do you wish to auto unseal vault(y/n)?y
Enter the key1:
Enter the key2:
Enter the key3:
Success! Uploaded policy: mcctb-policy
Error enabling approle auth: Error making API request.
URL: POST http://127.0.0.1:8200/v1/sys/auth/approle
Code: 400. Errors:
* path is already in use at approle/
Success! Enabled the kv secrets engine at: mcctb/
Success! Data written to: auth/approle/role/mcctb-app
Password updated successfully in the vault.
Synchronizing state of netapp-metrocluster-tiebreaker-
software.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable netapp-
metrocluster-tiebreaker-software
Created symlink /etc/systemd/system/multi-
user.target.wants/netapp-metrocluster-tiebreaker-software.service
→ /etc/systemd/system/netapp-metrocluster-tiebreaker-
software.service.
Attempting to start NetApp MetroCluster Tiebreaker software
services
Started NetApp MetroCluster Tiebreaker software services
Successfully installed NetApp MetroCluster Tiebreaker software
version 1.5.
```

升級現有的安裝

a. 確認已安裝受支援版本的OpenJDK、而且是主機上目前的Java版本。

> 若要升級至Tiebreaker 1.5、您必須安裝OpenJDK 17、18或19版。

```
[root@mcctb ~]# readlink -f /usr/bin/java
/usr/lib/jvm/java-19-openjdk-19.0.0.0.36-
2.rolling.el8.x86_64/bin/java
```

b. 確認Vault服務未密封且正在執行： `vault status`

```
[root@mcctb ~]# vault status
Key             Value
---             -----
Seal Type       shamir
Initialized     true
Sealed          false
Total Shares    5
Threshold       3
Version         1.12.2
Build Date      2022-11-23T12:53:46Z
Storage Type    file
Cluster Name    vault
Cluster ID      <cluster_id>
HA Enabled      false
```

c. 升級Tiebreaker軟體。

```
[root@mcctb ~]# rpm -Uvh NetApp-MetroCluster-Tiebreaker-Software-
1.5-1.x86_64.rpm
```

系統會顯示下列輸出以成功升級：

```
Verifying...
################################# [100%]
Preparing...
################################# [100%]
Updating / installing...
   1:NetApp-MetroCluster-Tiebreaker-
So################################# [ 50%]

Enter the absolute path for Java : /usr/lib/jvm/java-19-openjdk-
19.0.0.0.36-2.rolling.el8.x86_64/bin/java
Verifying if Java exists...
Found Java. Proceeding with the installation.
Enter host user account to use for the installation:
mcctbuser1
User account mcctbuser1 found. Proceeding with the installation
Sealed          false
Do you wish to auto unseal vault(y/n)?y
Enter the key1:
Enter the key2:
Enter the key3:
Success! Uploaded policy: mcctb-policy
Error enabling approle auth: Error making API request.
URL: POST http://127.0.0.1:8200/v1/sys/auth/approle
Code: 400. Errors:
* path is already in use at approle/
Success! Enabled the kv secrets engine at: mcctb/
Success! Data written to: auth/approle/role/mcctb-app
Enter database user name : root
Please enter database password for root
Enter password:
Password updated successfully in the database.
Password updated successfully in the vault.
Synchronizing state of netapp-metrocluster-tiebreaker-
software.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable netapp-
metrocluster-tiebreaker-software
Attempting to start NetApp MetroCluster Tiebreaker software
services
Started NetApp MetroCluster Tiebreaker software services
Successfully upgraded NetApp MetroCluster Tiebreaker software to
version 1.5.
Cleaning up / removing...
   2:NetApp-MetroCluster-Tiebreaker-
So################################# [100%]
```

> 如果您輸入錯誤的MySQL根密碼、則Tiebreaker軟體會指出已成功安裝、但會顯示「拒絕存取」訊息。若要解決此問題、您必須使用「rpm -e」命令解除安裝Tiebreaker軟體、然後使用正確的MySQL root密碼重新安裝軟體。

7. 開啟從斷路器主機到每個節點管理生命期和叢集管理生命期的SSH連線、以檢查斷路器與MetroCluster 該軟件的連線。

相關資訊

"NetApp支援"

版權資訊

Copyright © 2024 NetApp, Inc. 版權所有。台灣印製。非經版權所有人事先書面同意，不得將本受版權保護文件的任何部分以任何形式或任何方法（圖形、電子或機械）重製，包括影印、錄影、錄音或儲存至電子檢索系統中。

由 NetApp 版權資料衍伸之軟體必須遵守下列授權和免責聲明：

此軟體以 NETAPP「原樣」提供，不含任何明示或暗示的擔保，包括但不限於有關適售性或特定目的適用性之擔保，特此聲明。於任何情況下，就任何已造成或基於任何理論上責任之直接性、間接性、附隨性、特殊性、懲罰性或衍生性損害（包括但不限於替代商品或服務之採購；使用、資料或利潤上的損失；或企業營運中斷），無論是在使用此軟體時以任何方式所產生的契約、嚴格責任或侵權行為（包括疏忽或其他）等方面，NetApp 概不負責，即使已被告知有前述損害存在之可能性亦然。

NetApp 保留隨時變更本文所述之任何產品的權利，恕不另行通知。NetApp 不承擔因使用本文所述之產品而產生的責任或義務，除非明確經過 NetApp 書面同意。使用或購買此產品並不會在依據任何專利權、商標權或任何其他 NetApp 智慧財產權的情況下轉讓授權。

本手冊所述之產品受到一項（含）以上的美國專利、國外專利或申請中專利所保障。

有限權利說明：政府機關的使用、複製或公開揭露須受 DFARS 252.227-7013（2014 年 2 月）和 FAR 52.227-19（2007 年 12 月）中的「技術資料權利 - 非商業項目」條款 (b)(3) 小段所述之限制。

此處所含屬於商業產品和 / 或商業服務（如 FAR 2.101 所定義）的資料均為 NetApp, Inc. 所有。根據本協議提供的所有 NetApp 技術資料和電腦軟體皆屬於商業性質，並且完全由私人出資開發。 美國政府對於該資料具有非專屬、非轉讓、非轉授權、全球性、有限且不可撤銷的使用權限，僅限於美國政府為傳輸此資料所訂合約所允許之範圍，並基於履行該合約之目的方可使用。除非本文另有規定，否則未經 NetApp Inc. 事前書面許可，不得逕行使用、揭露、重製、修改、履行或展示該資料。美國政府授予國防部之許可權利，僅適用於 DFARS 條款 252.227-7015(b)（2014 年 2 月）所述權利。

商標資訊

NETAPP、NETAPP 標誌及 http://www.netapp.com/TM 所列之標章均為 NetApp, Inc. 的商標。文中所涉及的所有其他公司或產品名稱，均為其各自所有者的商標，不得侵犯。